

Topic 05: First-Order Linear Equations & The Integrating Factor Method

Professor Leonard Master Series · Lessons 15 to 18

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Product Rule in Reverse

"Have you ever looked at an expression and thought: 'Man, that looks ALMOST like the product rule, but something is missing'?"

Consider the product rule from Calc 1:

$$\frac{d}{dx}[\mu(x)y(x)] = \mu(x)\frac{dy}{dx} + \mu'(x)y$$

Now look at a standard linear differential equation:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

If we multiply the entire equation by a magic function $\mu(x)$:

$$\mu(x)\frac{dy}{dx} + \mu(x)P(x)y = \mu(x)Q(x)$$

We want the left-hand side to match the product rule exactly! That requires:

$$\mu'(x) = \mu(x)P(x) \implies \frac{d\mu}{\mu} = P(x)dx \implies \ln|\mu| = \int P(x)dx \implies \mu(x) = e^{\int P(x)dx}$$

That is the entire secret! $\mu(x)$ is not a magic trick; it is simply the exact factor needed to force the left side into the derivative of a single product!

2. Leonard's 5-Step Protocol

1. **Standard Form:** Make sure the coefficient of y' is **1**:

$$y' + P(x)y = Q(x)$$

2. **Find $\mu(x)$:**

$$\mu(x) = e^{\int P(x)dx}$$

3. **Multiply Through:**

$$\mu(x)y' + \mu(x)P(x)y = \mu(x)Q(x)$$

4. Condense Left Side:

$$\frac{d}{dx}[\mu(x)y] = \mu(x)Q(x)$$

5. Integrate and Divide:

$$\mu(x)y = \int \mu(x)Q(x)dx + C \implies y(x) = \frac{1}{\mu(x)} \left[\int \mu(x)Q(x)dx + C \right]$$

3. Fully Worked Master Example

Problem: Solve $x \frac{dy}{dx} - 3y = x^4 e^x$, $x > 0$.

- **Step 1: Put in standard form:** Divide every term by x :

$$\frac{dy}{dx} - \frac{3}{x}y = x^3 e^x$$

Here, $P(x) = -\frac{3}{x}$ and $Q(x) = x^3 e^x$.

- **Step 2: Calculate $\mu(x)$:**

$$\int P(x)dx = \int -\frac{3}{x}dx = -3 \ln x = \ln(x^{-3})$$

$$\mu(x) = e^{\ln(x^{-3})} = x^{-3} = \frac{1}{x^3}$$

- **Step 3: Multiply through & condense:**

$$\frac{1}{x^3} \frac{dy}{dx} - \frac{3}{x^4}y = \frac{1}{x^3}(x^3 e^x)$$

$$\frac{d}{dx} \left[\frac{1}{x^3}y \right] = e^x$$

- **Step 4: Integrate both sides:**

$$\frac{1}{x^3}y = \int e^x dx = e^x + C$$

- **Step 5: Multiply by x^3 :**

$$y(x) = x^3 e^x + Cx^3$$

4. Leonard's Red Flag Alerts

-  **Red Flag 1: Forgetting the Minus Sign in $P(x)$:** If your equation is $y' - \frac{3}{x}y = \dots$, $P(x) = -3/x$, NOT $+3/x$. Dropping the negative sign inverts $\mu(x)$ and ruins the problem!

-  **Red Flag 2: Distributing C :** Notice that $y = x^3e^x + Cx^3$. The $+C$ gets multiplied by x^3 too! Writing $y = x^3e^x + C$ is a fatal error.